Instrumental Variables and Two-Stage Least Squares

02 Nov 2026 | Chapter 12

ECO 351 – Introduction to Econometrics

Shi Qiu – Fall 2026

Recall: the source of a price change

Last class, a demand shift and a tax-induced supply shift both changed market price.

1. Which shift can move observations along a fixed demand curve?
2. Why could a tax chosen in response to anti-smoking attitudes fail as an instrument?

Explain the economic channel before naming a statistical test.

Demand shocks and the structural equation

For 48 US states in 1995, let Q_i be per-capita cigarette sales and P_i the real price.

$$\ln Q_i = \beta_0 + \beta_1 \ln P_i + u_i.$$

Here u_i includes omitted demand determinants. A demand shock can raise both Q_i and the equilibrium P_i, so

$$\mathrm{Cov}(\ln P_i, u_i) \neq 0.$$

A supply-side tax shift offers a different comparison, provided it leaves the omitted demand determinants unchanged.

The conditions for a valid instrument

Let $X = \ln P$, $Y = \ln Q$, and Z be the real general-sales-tax component per pack.

Condition	Meaning in the demand example
Relevance	$\text{Cov}(Z, X) \neq 0$: tax predicts price.
Exogeneity	$\text{Cov}(Z, u) = 0$: tax is unrelated to omitted demand determinants.
Exclusion	Tax affects demand through price, with no additional direct demand channel.

The exclusion argument is part of specifying the structural equation and defending the error-orthogonality condition.

Discussion: a strong tax-price relationship

A tax strongly predicts cigarette prices. States that dislike smoking also choose higher taxes and adopt smoke-free laws. Which condition is supported by the price regression? Which condition remains threatened?

Debrief: evidence and the remaining assumption

The price regression supports relevance. Attitudes and accompanying laws may change demand directly and correlate with the tax, threatening instrument exogeneity or exclusion.

First stage: tax and price

$$X_i = \pi_0 + \pi_1 Z_i + v_i.$$

π_1 measures the tax-price relationship. The data give:

```
. regress logp salestax, vce(robust)
Linear regression                               Number of obs     =         48
------------------------------------------------------------------------------
             |               Robust
        logp | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    salestax |   .0307289   .0048354     6.35   0.000     .0209956    .0404621
       _cons |   4.616546   .0289177   159.64   0.000     4.558338    4.674755
------------------------------------------------------------------------------
```

Which row provides evidence of relevance?

Reduced form: tax and sales

$$Y_i = \rho_0 + \rho_1 Z_i + e_i.$$

ρ_1 measures the tax-sales relationship:

```
. regress logq salestax, vce(robust)
Linear regression                               Number of obs     =         48
------------------------------------------------------------------------------
             |               Robust
        logq | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    salestax |  -.0332974   .0102541    -3.25   0.002    -.0539378    -.012657
       _cons |   4.717449   .0650586    72.51   0.000     4.586493    4.848405
------------------------------------------------------------------------------
```

Under the instrument restrictions, $\rho_1 = \beta_1 \pi_1$.

The IV ratio

With one endogenous regressor and one instrument,

$$\beta_1 = \frac{\rho_1}{\pi_1}, \qquad \widehat{\beta}_{IV} = \frac{\widehat{\rho}_1}{\widehat{\pi}_1}.$$

Using the two estimated tax relationships,

$$\widehat{\beta}_{IV} = \frac{-0.0332974}{0.0307289} \approx -1.084.$$

Under the maintained model and instrument restrictions, a 1% tax-induced price increase implies approximately a 1.084% decrease in quantity demanded.

Exercise: a different proposed instrument

In a constructed market example, an external production-cost index predicts a 0.02 increase in log price and a 0.03 decrease in log sales per index unit.

1. Calculate the implied IV demand elasticity.
2. Give the approximate quantity response to a 5% price increase.
3. The cost change also lowers product quality. Which identifying argument now needs attention?

Debrief: the ratio and the quality channel

$$\widehat{\beta}_{IV} = -0.03/0.02 = -1.5, \qquad \Delta Q\% \approx (-1.5)(5\%) = -7.5\%.$$

The quality change supplies a second route from the proposed instrument to demand. The price-only exclusion argument therefore fails unless the specification and assumptions address that route.

Correct arithmetic cannot compensate for an invalid source of price variation.

Two-stage least squares

Stage 1: estimate the price equation and retain fitted log price.

$$\widehat{X}_i = \widehat{\pi}_0 + \widehat{\pi}_1 Z_i.$$

Stage 2: regress log quantity on that fitted component.

$$Y_i = b_0 + b_1 \widehat{X}_i + \text{second-stage residual}.$$

The resulting slope equals the simple IV ratio. For reported standard errors, use Stata's dedicated IV command.

Stata: the IV estimate and uncertainty

```
. ivregress 2sls logq (logp = salestax), vce(robust)
Instrumental-variables 2SLS regression            Number of obs   =         48
------------------------------------------------------------------------------
             |               Robust
        logq | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        logp |  -1.083587   .3122036    -3.47   0.001    -1.695495    -.471679
       _cons |   9.719877   1.496143     6.50   0.000     6.78749    12.65226
------------------------------------------------------------------------------
Endogenous: logp
Exogenous:  salestax
```

`logq` is the outcome. Inside parentheses, `logp` is endogenous and `salestax` is the excluded instrument.

The estimated price elasticity is -1.084, with 95% interval approximately $[-1.695, -0.472]$.

OLS and IV use different price comparisons

The same 48-state data give

Estimator	Price coefficient	Robust SE
OLS	-1.213	0.195
IV using sales tax	-1.084	0.312

OLS uses all price variation. IV uses the price variation associated with the proposed instrument.

A larger IV standard error does not make OLS unbiased. A difference between the point estimates does not prove that the instrument is valid.

An included control belongs in both stages

Let W_i be log real per-capita income, treated as an exogenous control.

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + u_i,$$
$$X_i = \pi_0 + \pi_1 Z_i + \pi_2 W_i + v_i.$$

Stage 2 includes fitted X_i and the same W_i.

We now require the tax to provide relevant, valid price variation *conditional on income*. Including income does not establish its exogeneity.

Stata: IV with income held fixed

```
. ivregress 2sls logq logincome (logp = salestax), vce(robust)
Instrumental-variables 2SLS regression            Number of obs   =         48
------------------------------------------------------------------------------
             |               Robust
        logq | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        logp |  -1.143375   .3604805    -3.17   0.002    -1.849904   -.4368463
   logincome |   .2145153   .3018477     0.71   0.477    -.3770953    .8061258
       _cons |   9.430658   1.219402     7.73   0.000     7.040675    11.82064
------------------------------------------------------------------------------
Endogenous: logp
Exogenous:  logincome salestax
```

The fitted price elasticity is about -1.143, conditional on log income and the maintained instrument assumptions.

First-stage evidence conditional on income

```
. estat firststage
  First-stage regression summary statistics
  ------------------------------------------------------------------------------
               |             Adjusted      Partial       Robust
      Variable |   R-sq.       R-sq.        R-sq.       F(1,45)   Prob > F
  -------------+----------------------------------------------------------------
          logp |  0.6389      0.6228       0.5009       44.7305    0.0000
  ------------------------------------------------------------------------------
```

The excluded-instrument robust $F(1, 45)$ is about 44.73.

This is evidence about relevance. It cannot establish exclusion or exogeneity. Weak instruments can make 2SLS unstable and routine confidence intervals unreliable.

Discussion: can the first stage settle the causal claim?

A researcher writes:

> *The first-stage F is 44.73, so the tax must be unrelated to omitted demand shocks. Our IV estimate is therefore causal.*

Which part of that argument is unsupported? Give one concrete feature of state tax policy that you would investigate.

Debrief: the identifying argument

A strong tax-price relationship supports relevance. The demand interpretation still requires an argument about tax policy and omitted demand determinants.

Higher-tax states might also impose smoking restrictions or have stronger anti-smoking preferences. Either can undermine the tax instrument in this specification.

Statistical precision and instrument credibility answer different questions.

IV in an applied project

Before using IV, state the endogenous regressor, the excluded instrument, and the economic argument for its validity. Then examine the first stage and report the IV estimate with uncertainty.

IV is available as an optional project method. A carefully justified regression remains preferable to an unconvincing instrument.

Next class: repeated observations offer another way to address confounding that stays constant within a place over time.

Midterm 2 retains its announced coverage. IV is outside that coverage.